

Aishwarya Ganesan

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PROFESSIONAL SUMMARY

AI/ML engineer with 4 years of production software engineering experience and an **MSc in Machine Learning from KTH**. Builds **production AI systems** spanning entity resolution, LLM-assisted information extraction, semantic retrieval, and ML-powered backend services. Experienced in designing **scalable data pipelines** processing hundreds of thousands of product listings, combining deterministic algorithms, embeddings, and large language models to build reliable intelligent systems.

EXPERIENCE

AI/ML Engineer · Freelance, India | 2025 – Present

- Architected and implemented an **AI-assisted entity-resolution pipeline** for 350K+ electronics product listings across 8+ retailers, consolidating noisy supplier catalogs into canonical components for downstream search and analytics.
- Built a **hybrid matching engine** combining deterministic part-number extraction with embedding-based similarity search, balancing high-precision grouping with robust handling of incomplete and inconsistent product data.
- Developed **LLM-assisted extraction pipelines** using Claude API for structured product information extraction, category assignment, and metadata generation from heterogeneous supplier catalogs.
- Built a **hybrid retrieval system** using PostgreSQL, pgvector, and keyword search to improve product discovery through dense-vector retrieval and lexical ranking.
- Designed **modular data ingestion and normalization pipelines** to support scalable catalog processing, evaluation, and iterative improvements to retrieval quality.

ADAS Perception Research Engineer (Master Thesis) · Scania CV AB, Sweden | 2023 – 2024

- Developed a research ADAS prototype using a two-stage **Deep Reinforcement Learning (DRL)** pipeline to improve post-detection lane accuracy on curved, faded, or partially blocked lanes. [Dive Deeper →](#) | [Published Thesis \(DiVA\)](#)
- Handled the entire **data pipeline end-to-end**: dataset cleaning, ground-truth labeling (Scania and TuSimple datasets), augmentation, and benchmarking against segmentation and object-detection methods.
- Trained a **lane-refinement model** in PyTorch pairing a ResNet detector with a Double Deep Q-Network (Double DQN), reaching **89.2% accuracy**; a Bézier curve-control extension traded accuracy (78.3%) for a larger action space.
- Evaluated deployment viability through **model pruning and quantisation** for edge hardware, and end-to-end comparison with industry-standard lane-detection approaches, highlighting DRL's strengths in refinement but limitations in curve-based policies.

Senior Software Engineer · Bosch Global Software Technologies, India | 2017 – 2021

- Led **Electronic Stability Program (ESP) function development** (Emergency Brake Assist, Cruise, Boost, Hold) across 7+ production programs for Fiat and Iveco, spanning passenger and light commercial vehicles in the EU, Brazil, and China.
- Adapted modular control logic to diverse OEM specifications, enabling synchronized rollouts and **reducing code rework by 75%**.
- Achieved **ASPICE Level 3** certification across two production platforms by refactoring legacy control modules and improving traceability, contributing to a **100% audit pass rate**.
- Authored standardized engineering guidelines in Bosch's **Software Development Handbook (SDH)**, streamlining integration, change management, and delivery workflows, improving build consistency and cutting integration defects.

TECHNICAL SKILLS

Languages: Python, C++, SQL

AI: LLM Applications, Claude API, Embeddings, Vector Search, Hybrid Retrieval, Semantic Search, Entity Resolution

Machine Learning: PyTorch, TensorFlow, scikit-learn, NumPy, pandas, Reinforcement Learning, Model Quantization, Pruning

Backend & Data: FastAPI, PostgreSQL, pgvector, REST APIs, JSONB, Data Pipelines, Web Scraping, Playwright, Docker, GCP, Git

Computer Vision: OpenCV, ResNet, Vision Transformers (ViT), Masked Autoencoders, Grad-CAM

Engineering: System Design, ML Evaluation, Error Analysis, Modular Architecture, Debugging

Embedded: Embedded C/C++, ESP, ASPICE

SELECTED PROJECTS

Deep Learning Model Extraction Attacks | PyTorch, ResNet-34/50, CIFAR-10/100

- Demonstrated the viability of black-box model extraction attacks by training a ResNet-34 surrogate to replicate a ResNet-50 victim using 25k API queries, achieving 82.88% top-1 agreement on CIFAR. [Dive Deeper →](#)
- Ran ablation studies on softmax access, query strategy, and architecture mismatch, finding low adversarial transfer (under 3.5%) but clear training-data leakage on out-of-distribution inputs.

Masked Autoencoders for Vision Representation Learning | PyTorch, ViT-B/16, Grad-CAM, Imagenette2-320

- Replicated the MAE architecture (ViT-B/16 encoder, shallow decoder) pre-trained on Imagenette2-320 with 75% masking; achieved 66.85% top-1 in linear probing. [Dive Deeper →](#)
- Applied Grad-CAM to assess semantic focus; found meaningful object parts despite blurry edges, and that disabling augmentation improved accuracy by 1.5%.

EDUCATION

KTH - Royal Institute of Technology, Sweden

M.Sc. in Machine Learning (ML) | GPA: 4.69/5.0 | **KTH One-Year Scholarship for Academic Excellence**

Amrita Vishwa Vidyapeetham, India

B.Tech in Electronics and Communication Engineering (ECE) | CGPA: 8.92/10.0 | **First Class with Distinction**